

A Lightweight and Explainable Deep Learning Framework for Phonocardiogram Classification: A Comparative Study Across Architectures and Features with Edge-Deployment Perspective

Md Toufiqu Rahman, Bo Yuan

Course Project, INTRO DEEP LEARNING 16:332:530:01, Department of Electrical and Electronic Engineering
Rutgers University
New Jersey, USA
mr2103@rutgers.edu

Abstract—Cardiovascular diseases (CVDs) remain a leading cause of global mortality, and phonocardiogram (PCG) analysis offers a non-invasive avenue for early detection of valvular heart diseases (VHDs). Recent deep-learning approaches achieve high accuracy but rely on large architectures that are unsuitable for edge-device deployment. This work presents a systematic comparative study of seven deep architectures (CNN-1/2, SepCNN-1/2, Conv-LSTM, Conv-BiLSTM, Conv-GRU) across three classical feature representations (DWT scalogram, MFCC, log-mel spectrogram) on the Open-Access Heart Sound (OAHS) five-class VHD task. In addition, we propose TinyPCGNet, an ultra-tiny dual-branch fusion model combining MFCC and log-mel spectrograms through depthwise-separable convolutions and squeeze-and-excite attention. Across 22 controlled experiments using identical training protocols and stratified 5-fold cross-validation, we find that recurrent models on MFCC achieve the highest accuracy (Conv-BiLSTM $\times$ MFCC: 99.5%, 100K parameters) while TinyPCGNet achieves 92.8% with only 5.1K parameters — a $20\times$ parameter reduction at a 6.7% accuracy cost. These results suggest that the proposed model can be implemented on edge devices to provide an initial review or screening of PCG signals before further clinical assessment. Although identifying the optimal trade-off between accuracy and model complexity remains an open question, this work provides a practical framework for designing lightweight PCG classification models that balance diagnostic performance with deployability on resource-constrained hardware. Also, Grad-CAM analysis reveals class-discriminative temporal attention patterns. The results characterize the parameter-versus-accuracy trade-off frontier for PCG classification and demonstrate that ultra-lightweight models remain viable for resource-constrained edge devices.

Index Terms—phonocardiogram, deep learning, lightweight models, edge AI, explainable AI, Grad-CAM, valvular heart disease, depthwise-separable convolution

I. INTRODUCTION

Cardiovascular diseases account for an estimated 17.9 million deaths each year worldwide, with valvular heart diseases (VHDs) such as aortic stenosis (AS), mitral regurgitation (MR), mitral stenosis (MS), and mitral valve prolapse (MVP) representing a significant fraction of clinically relevant cardiac pathologies. Cardiac auscultation, the practice of listening to

heart sounds through a stethoscope, remains the first-line non-invasive diagnostic for VHDs in most clinical settings. Yet manual auscultation is highly subjective: studies report inter-rater agreement as low as 20–40% among general practitioners, and access to expert cardiologists is severely limited in low-resource settings.

Phonocardiogram (PCG) signals — digital recordings of heart sounds — enable automated, reproducible analysis of cardiac acoustic signatures. Over the past five years, deep learning has substantially advanced automated PCG classification. Notable contributions include SpectroCardioNet [1], CNN-LSTM with Grad-CAM explainability [5], SAR-CardioNet [2], the explainable 2D-CNN of Bhardwaj *et al.* [4], and LtXe-EnsNet [3], which combines MFCC and spectrogram features through an attention-based ensemble.

Despite these advances, two limitations persist. First, most existing models are computationally heavy, with parameter counts in the 10^5 – 10^6 range, making real-time edge deployment on microcontrollers (MCUs) or always-on wearables impractical. Second, the broader *parameter-versus-accuracy trade-off* has not been systematically characterized: each paper proposes a single model and tunes it heavily, making direct architectural comparisons difficult.

This work makes three contributions:

- 1) A **controlled 22-experiment comparison** of seven deep architectures across three feature representations using identical training protocols and fold splits, enabling apples-to-apples architectural comparison.
- 2) **TinyPCGNet**, a 5.1K-parameter dual-branch fusion model that achieves 92.8% accuracy — a $20\times$ parameter reduction over the state-of-the-art LtXe-EnsNet [3] at a modest accuracy cost.
- 3) A **Grad-CAM analysis** of TinyPCGNet showing class-discriminative attention patterns consistent with the cardiac-cycle structure of each VHD type.

The remainder of the paper is organized as follows. Section II motivates the edge-deployment perspective. Section III describes the proposed system pipeline. Sections IV and V

describe the dataset and feature extraction. Section VI presents the model architectures. Section VII details the training protocol. Section VIII reports numerical results and Grad-CAM analysis. Section IX discusses implications. Section X outlines future directions, and Section XI concludes.

II. MOTIVATION

A. Why edge deployment matters

Three practical considerations motivate the development of lightweight PCG classifiers.

Hardware constraints. Cortex-M class microcontrollers commonly used in wearable cardiac monitors offer less than 512 KB of RAM and under 2 MB of flash memory. Even a Raspberry Pi Zero, while substantially more capable, has limited inference budgets for always-on monitoring. Models in the 10^5 – 10^6 parameter range cannot fit on MCUs without aggressive quantization, and even then incur latency that precludes real-time use.

Battery life. Energy per inference scales approximately with FLOPs. A model requiring 4×10^7 FLOPs per inference will drain a coin-cell battery substantially faster than a model requiring 4×10^6 FLOPs. For wearable devices intended to provide continuous cardiac monitoring, the $10\times$ difference is the difference between hours and days of operation.

Privacy and connectivity. On-device inference preserves patient privacy by avoiding the transmission of acoustic recordings to cloud servers, and ensures system reliability in settings with intermittent connectivity — including the rural and low-resource contexts where automated screening is most valuable.

B. Why a controlled comparison is needed

A casual reading of the literature suggests that “deeper and more complex always wins.” However, individual papers each propose one model trained with their own preferred hyperparameters, making cross-paper comparisons unreliable. Several questions cannot be answered from this literature:

- Does fusion of multiple features always outperform a single well-chosen feature?
- Are depthwise-separable convolutions worth the accuracy cost in this domain?
- Do recurrent layers add value on top of convolutional backbones?
- What is the actual Pareto frontier of accuracy versus parameters?

This work answers these questions through systematic experimentation.

III. SYSTEM MODEL

Figure 1 shows the complete classification pipeline. A raw PCG recording (typically captured by a digital stethoscope) is preprocessed, transformed into one of three time-frequency feature representations, and passed through a deep classifier to produce a softmax probability over five classes. Section VIII-E additionally applies Grad-CAM on the classifier’s last convolutional layer to produce visual explanations.

The key design choice is that all three feature representations share the same time axis (94 frames), so a single model architecture can be applied to any feature with no architectural changes. This enables the controlled comparison reported in Section VIII.

IV. DATASET

We use the **Open-Access Heart Sound (OAHS) dataset** [6], a balanced, noise-free corpus of 1000 PCG recordings covering five classes:

- **AS** — Aortic Stenosis: a systolic murmur caused by obstructed flow through a stenotic aortic valve, typically presenting with a mid-systolic crescendo-decrescendo shape.
- **MR** — Mitral Regurgitation: a holosystolic murmur from backward flow through an incompetent mitral valve.
- **MS** — Mitral Stenosis: a low-frequency, mid-diastolic rumble caused by obstructed flow into the left ventricle.
- **MVP** — Mitral Valve Prolapse: a characteristic mid-to-late systolic click followed by a late-systolic murmur.
- **N** — Normal: regular S1 and S2 sounds with quiet systolic and diastolic intervals.

Each class contains 200 recordings, sampled at 8 kHz, and known to be free of significant ambient noise. We chose OAHS for three reasons: (a) it is class-balanced, eliminating class imbalance as a confounding factor; (b) its clean recordings allow us to compare model capacity without noise robustness as a confound; and (c) it is the most widely used benchmark for multi-class VHD classification in recent literature [1]–[5], enabling direct cross-paper comparison. Figure 2 shows one example 3-second waveform from each class, illustrating these distinct temporal signatures.

Validation protocol. We employ stratified 5-fold cross-validation (seed = 42) with the fold splits saved to disk and **reused identically across all 22 experiments**, including TinyPCGNet. This guarantees apples-to-apples comparison: every experiment is evaluated on exactly the same held-out samples in each fold.

V. DATA PROCESSING AND FEATURE EXTRACTION

A. Preprocessing chain

Each recording is processed through a four-stage chain:

- 1) **Resampling** to 2 kHz, sufficient for VHD information which lies almost entirely below 1 kHz.
- 2) **Length normalization**: each recording is truncated or zero-padded to exactly 3 seconds (6000 samples).
- 3) **RMS normalization**: $x \leftarrow x/\|x\|_2$ to eliminate microphone-gain differences.
- 4) **Z-score normalization**: $x \leftarrow (x - \mu_x)/\sigma_x$ to zero-mean, unit-variance the waveform.

We do **not** apply an explicit bandpass filter. The mel filterbank used in MFCC and log-mel features performs implicit frequency selection (20–1000 Hz), and bandpass filters introduce phase distortion that can mislead CNNs.

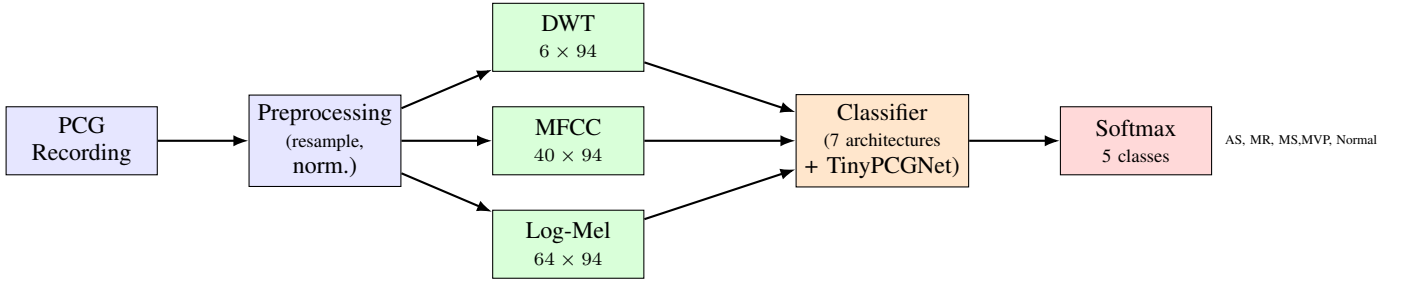


Fig. 1: End-to-end PCG classification pipeline. A 3-second PCG recording is preprocessed and converted to one of three feature representations (DWT scalogram, MFCC, or log-mel spectrogram, each of shape rows $\times$ 94), then fed into one of eight classifier variants (seven baselines and the proposed TinyPCGNet). The output is a softmax probability over the five classes: AS, MR, MS, MVP, and Normal.

B. Three feature representations

All three features are computed on the cleaned waveform and yield arrays of shape rows $\times$ 94 $\times$ 1, where 94 is the number of time frames over the 3-second segment.

1) *Log-mel spectrogram* ($64 \times 94 \times 1$): The Short-Time Fourier Transform (STFT) is computed with FFT size 512, hop length 64, and a Hann window. The power spectrum is passed through a 64-band mel filterbank with cutoffs at 20 Hz and 1000 Hz, then converted to decibels and z-score normalized.

2) *MFCC* ($40 \times 94 \times 1$): 40 Mel-Frequency Cepstral Coefficients are computed per frame using the same FFT/hop settings. MFCC captures the spectral envelope shape, which is particularly useful for distinguishing murmur types that share the same time location but differ in spectral character.

3) *DWT scalogram* ($6 \times 94 \times 1$): A 5-level Daubechies-4 (db4) discrete wavelet decomposition yields one approximation coefficient array (cA_5) and five detail coefficient arrays (cD_5 through cD_1). Each is FFT-resampled to a common length of 94, and the six arrays are stacked vertically with the coarsest approximation at the top.

VI. MODEL ARCHITECTURES

We evaluate seven baseline architectures and the proposed TinyPCGNet. All baselines share a common conv-block template: $\text{Conv2D}(32, 3 \times 3) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{MaxPool}(2 \times 2)$, with SeparableConv2D substituted in the separable variants.

A. The seven baselines

- **CNN-1 / CNN-2:** one or two standard conv blocks followed by Global Average Pooling (GAP), Dropout(0.3), and a softmax dense layer.
- **SepCNN-1 / SepCNN-2:** same structure but with Depthwise-Separable Conv2D layers in place of standard Conv2D.
- **Conv-LSTM:** two conv blocks, followed by a permutation and reshape converting the (H', W', C) feature map into a $(W', H' \cdot C)$ sequence, then an LSTM(32) and softmax.
- **Conv-BiLSTM:** identical to Conv-LSTM but with a Bidirectional LSTM.
- **Conv-GRU:** identical to Conv-LSTM but with a GRU.

B. TinyPCGNet — the proposed model

TinyPCGNet (Fig. 4) is a dual-branch fusion model designed to combine the strengths of MFCC and log-mel spectrogram representations.

The architectural choices reflect three design principles:

- **Depthwise-separable convolutions** reduce parameter count by approximately $8\times$ relative to standard Conv2D for 3×3 kernels, with minimal accuracy impact at the depths used here.
- **Squeeze-and-Excite (SE) attention** [7] provides channel attention at < 200 parameters per block, replacing the heavier BiLSTM-attention used in LtXe-EnsNet.
- **Global Average Pooling** eliminates the parameter-heavy Dense layer that typically dominates parameter counts in comparable models, while serving as a structural regularizer.

VII. TRAINING PROTOCOL

To ensure the comparison is apples-to-apples, all 22 experiments share identical training settings. Only the architecture and the input feature vary.

A. Optimizer and learning-rate schedule

We use Adam with a base learning rate of 3×10^{-4} , preceded by a 5-epoch linear warmup from 5×10^{-5} . The warmup is crucial: without it, occasional folds diverge during the first few mini-batches when batch-normalization statistics are unstable.

After warmup, we apply ReduceLROnPlateau on validation loss (factor 0.5, patience 10), which adaptively halves the learning rate when progress stalls.

B. Early stopping and protection against premature termination

We use EarlyStopping on validation accuracy with patience 20, but protected against premature triggering by requiring a minimum of 25 epochs of training. This prevents EarlyStopping from killing models that have not yet escaped a bad initialization.

OAHS Dataset: Example 3-second Raw PCG Segments (5 Classes)

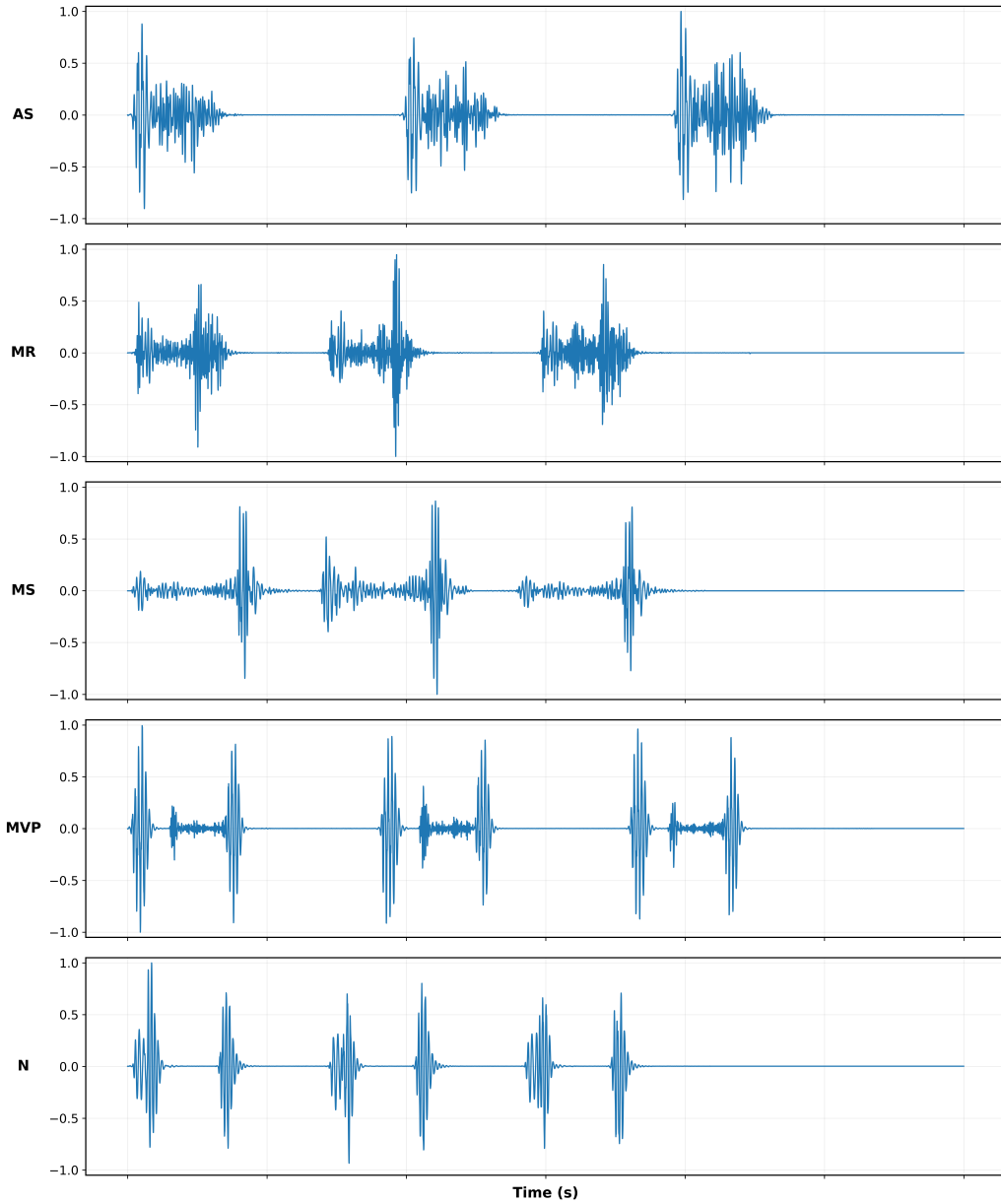


Fig. 2: Example 3-second raw PCG waveforms from the OAHS dataset, one per class. The five classes exhibit visibly different temporal signatures: AS shows mid-systolic crescendo-decrescendo bursts; MR exhibits broader holosystolic activity; MS displays low-amplitude diastolic rumbles between strong S1/S2 transients; MVP shows a characteristic mid-systolic click followed by a late-systolic murmur; and Normal (N) has distinct S1 and S2 sounds with quiet systolic and diastolic intervals.

C. Other settings

- Batch size: 32
- Maximum epochs: 100
- Loss: categorical cross-entropy
- Random seed: 42 for fold splits, fold-derived seeds for each training run
- Hardware: Google Colab T4 GPU

The complete code, including a retry-on-collapse mecha-

nism for TinyPCGNet that automatically re-trains a fold with a new seed if the first attempt converges below 50%, is publicly available.

VIII. RESULTS AND ANALYSIS

A. Headline ranking

Table I presents the complete 22-row ranking, sorted by mean cross-validation accuracy. Several findings emerge.

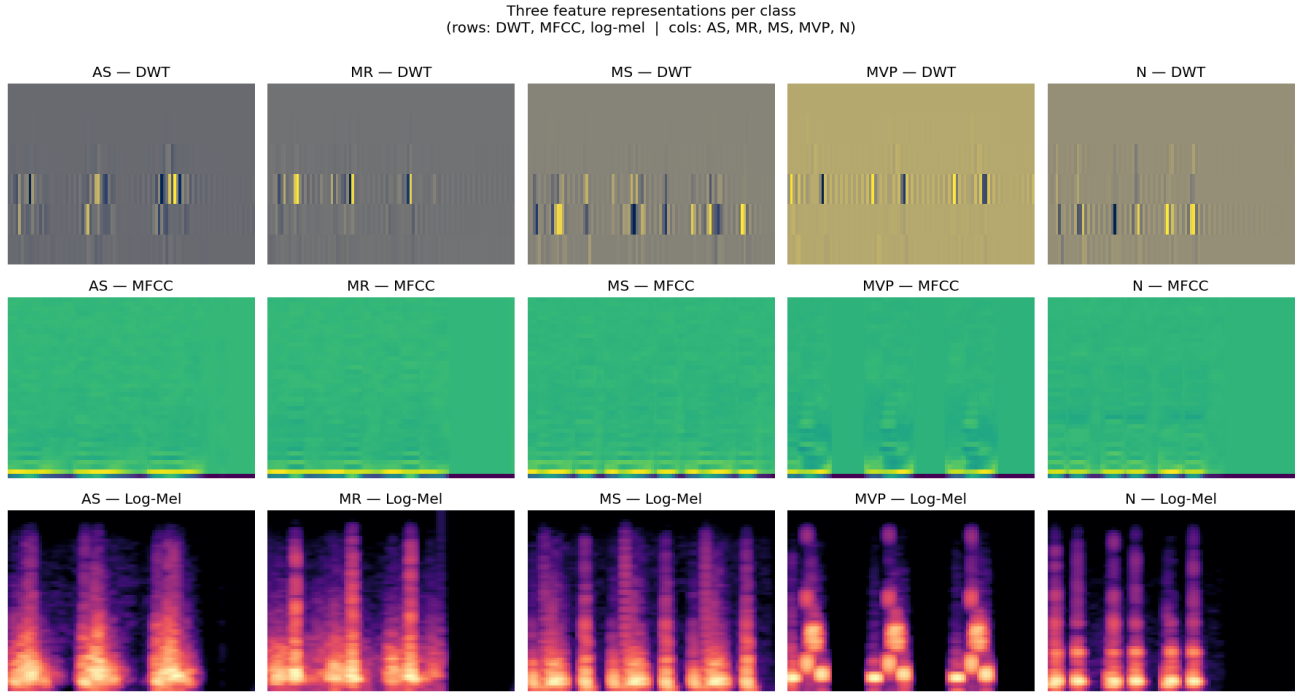


Fig. 3: Three feature representations per class (top row: DWT; middle row: MFCC; bottom row: log-mel). Each column shows a randomly chosen recording from one of the five classes. The five classes are distinguishable in the MFCC and log-mel representations but less so in the DWT scalogram, foreshadowing the results of Section VIII.

TABLE I: Full 22-experiment ranking on the OAHS five-class task, sorted by mean accuracy across stratified 5-fold cross-validation. The proposed TinyPCGNet (highlighted) achieves the smallest parameter count above 90% accuracy.

Rank	Model	Feature	Params (K)	FLOPs (M)	Accuracy (%)	F1 (%)
1	Conv-BiLSTM	MFCC	100.3	24.1	99.5	99.5
2	Conv-LSTM	MFCC	55.0	22.0	99.4	99.4
3	Conv-BiLSTM	LogMel	149.5	38.4	99.1	99.1
4	Conv-LSTM	LogMel	79.6	35.0	99.0	99.0
5	Conv-GRU	LogMel	62.2	34.2	98.9	98.9
6	Conv-GRU	MFCC	43.8	21.4	98.9	98.9
7	TinyPCGNet	MFCC+LogMel	5.1	4.3	92.8	92.8
8	CNN-2	MFCC	9.8	19.8	89.9	89.8
9	Conv-BiLSTM	DWT	34.8	4.1	88.6	88.5
10	Conv-LSTM	DWT	22.2	3.6	88.2	88.1
11	CNN-2	LogMel	9.8	31.7	87.7	87.6
12	Conv-GRU	DWT	19.2	3.4	83.3	83.1
13	SepCNN-2	MFCC	1.6	3.1	75.7	75.1
14	CNN-2	DWT	9.8	3.0	75.3	74.3
15	SepCNN-2	LogMel	1.6	4.9	72.7	71.6
16	CNN-1	MFCC	0.5	2.4	64.4	60.7
17	CNN-1	LogMel	0.5	3.9	59.5	54.3
18	SepCNN-2	DWT	1.6	0.5	59.2	54.2
19	CNN-1	DWT	0.5	0.4	50.2	44.6
20	SepCNN-1	MFCC	0.3	0.5	44.2	34.8
21	SepCNN-1	LogMel	0.3	0.9	33.0	21.1
22	SepCNN-1	DWT	0.3	0.1	24.3	16.3

Finding 1: Recurrent models dominate the top of the table. The top six positions are all occupied by Conv-LSTM, Conv-BiLSTM, or Conv-GRU on MFCC or log-mel features. Conv-BiLSTM $\times$ MFCC tops the ranking at 99.5% accuracy with 100K parameters, matching the reported accuracy of

LtXe-EnsNet [3] (141K parameters) and SpectroCardioNet [1] (193K parameters), while using fewer parameters.

Finding 2: MFCC outperforms log-mel which outperforms DWT. For every model architecture, MFCC and log-mel features substantially outperform DWT scalograms. The

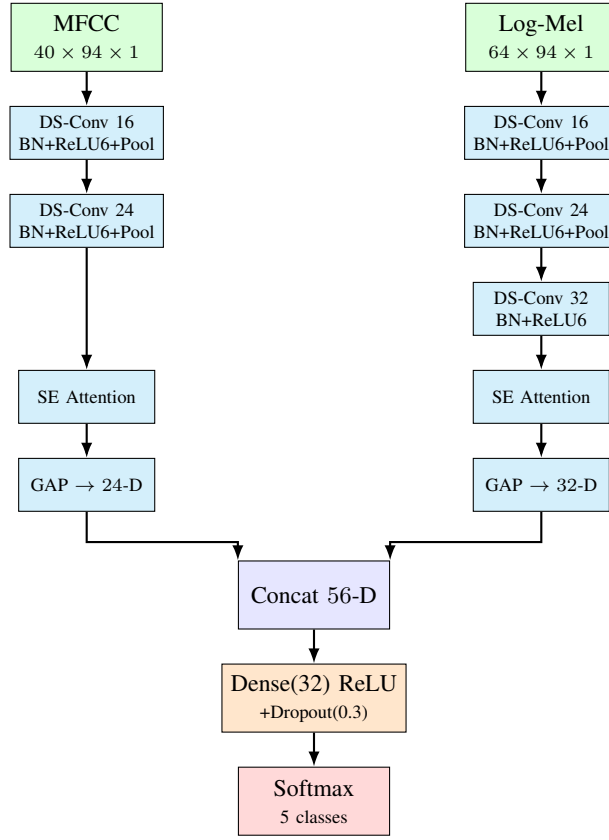


Fig. 4: TinyPCGNet v4.1 dual-branch architecture. The MFCC branch (left) processes 40 cepstral coefficients through two depthwise-separable convolutional blocks; the log-mel spectrogram branch (right) processes 64 mel bins through three blocks. Each branch then applies squeeze-and-excite channel attention and global average pooling, producing 24-D and 32-D feature vectors respectively. The concatenated 56-D representation is classified through a small dense head, for a total of ~ 5.1 K trainable parameters.

compactness and perceptual weighting of MFCC appear to capture VHD-discriminative information more efficiently than either the raw mel spectrogram or the wavelet decomposition.

Finding 3: TinyPCGNet sits at the Pareto knee. With 5.1K parameters, TinyPCGNet is the smallest model in the top half of the ranking and the smallest to exceed 90% accuracy. Models with fewer parameters (SepCNN-1 at 0.3K) collapse to near-random performance, identifying TinyPCGNet’s operating point as approximately the minimum-capacity solution to this task.

Finding 4: Separable convolutions underperform at this scale. SepCNN-2 (1.6K) achieves at most 76% versus CNN-2 (9.8K) at 90%. The $6\times$ parameter savings of separable convolutions cost more accuracy than they save when used in a shallow (≤ 2 -block) architecture. This contrasts with the deep-network setting (e.g., MobileNets [8]) where depthwise-separable convolutions shine in deep cascades.

B. The accuracy-versus-parameters trade-off

Figure 5 plots accuracy against parameter count (log-scale) and FLOPs for all 22 experiments. Three distinct operating regions emerge:

- **High-accuracy regime ($\geq 99\%$, 50–150K params).** Recurrent models on MFCC or log-mel. Suitable for

smartphone-class deployment.

- **Mid regime (90–98%, 5–50K params).** Contains TinyPCGNet at the lower bound. Suitable for Raspberry Pi or Cortex-A class devices.
- **Ultra-light regime ($< 90\%$, < 5 K params).** Accuracy drops sharply below 5K parameters. SepCNN-1 (0.3K) achieves only 24–44% accuracy, indicating insufficient capacity for the task.

The knee of the curve is at approximately 5K parameters, corresponding to TinyPCGNet, suggesting this is the minimum viable capacity for OAHS five-class classification.

C. Per-feature comparison

Figure 6 presents a model-by-feature heatmap of mean accuracy. The pattern is consistent across architectures: MFCC achieves the highest accuracy, log-mel a close second, and DWT trails by 5–15 percentage points across all models. Recurrent models extract considerably more value from MFCC than convolutional-only models do.

D. TinyPCGNet confusion matrix

Figure 7 shows the aggregate confusion matrix for TinyPCGNet, summed over all five folds. The Normal (N) class is near-perfectly classified, MVP and MS are well

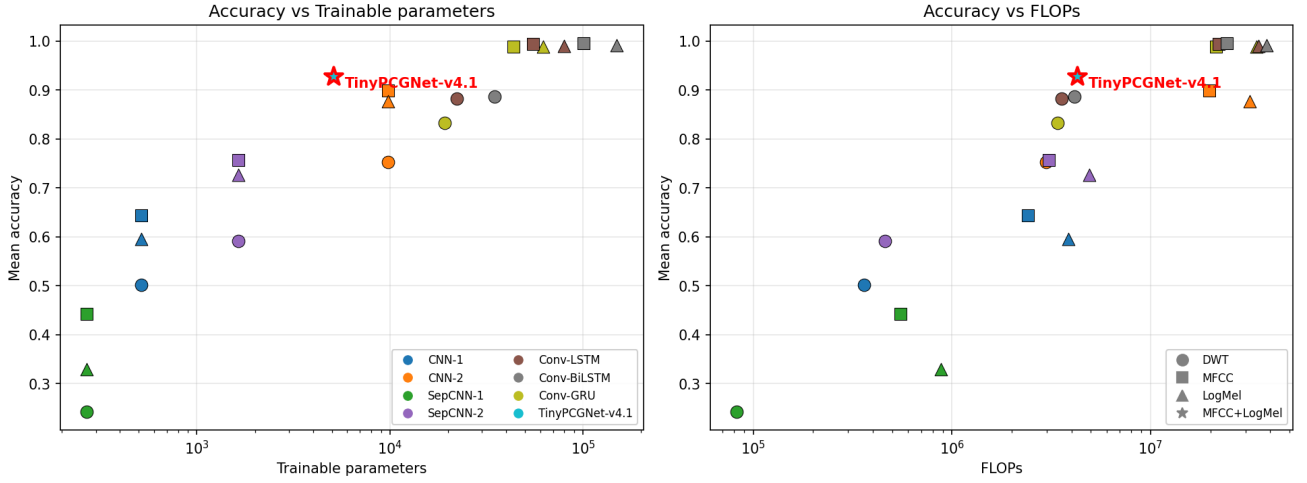


Fig. 5: Mean accuracy versus trainable parameter count (left, log scale) and FLOPs (right, log scale). Each marker represents one of the 22 experiments. Three operating regions are visible: a high-accuracy regime ($\geq 99\%$, 50–150K params), a mid regime (90–98%, 5–50K params), and an ultra-light regime ($< 90\%$, $< 5K$ params) where accuracy collapses sharply. TinyPCGNet (highlighted) sits at the knee of the curve.

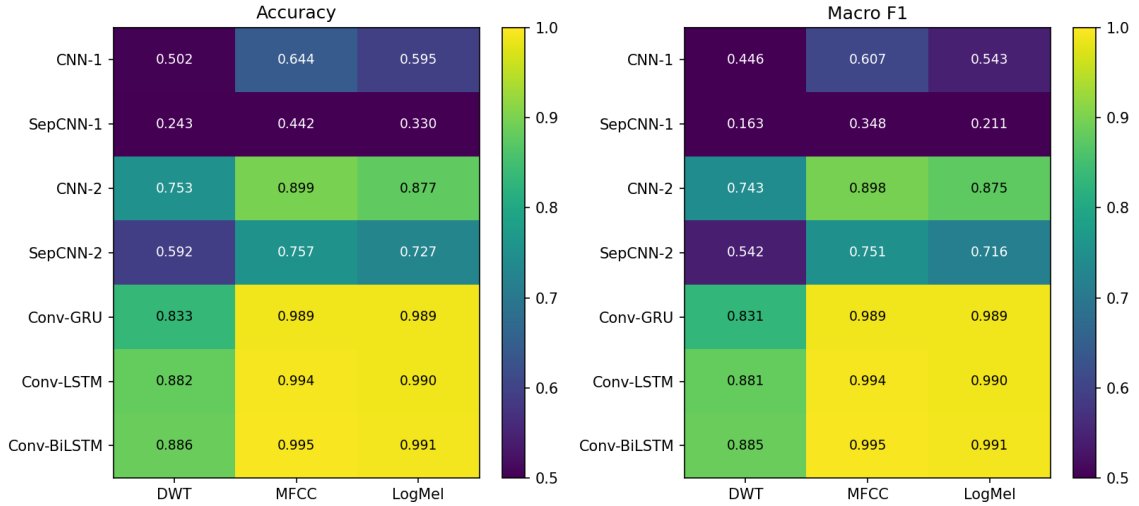


Fig. 6: Mean accuracy heatmap across model (rows) and feature (columns). Recurrent models on MFCC dominate; DWT consistently underperforms.

distinguished, but some confusion remains between AS and MR — both systolic murmurs that share temporal location but differ in spectral character.

E. Explainability with Grad-CAM

To produce visual explanations of TinyPCGNet’s decisions, we apply Grad-CAM [9] on the last separable convolutional layer of the spectrogram branch (`spec_b3_sepconv`). We perform four analyses.

1) *Per-class representative heatmaps*: For each class, we identify the highest-confidence correctly-classified sample in the held-out validation set and display the input spectrogram alongside its Grad-CAM overlay (Fig. 8). The heatmaps show clear class-discriminative patterns: AS attention concentrates on systolic windows, MR exhibits broader temporal coverage

consistent with holosystolic murmur, MS shows diastolic emphasis, and MVP shows a characteristic late-systolic peak.

2) *Correct versus incorrect predictions*: For classes with misclassifications, we display correct and incorrect predictions side-by-side (Fig. 9). Misclassifications often exhibit diffuse attention compared to confident correct predictions, suggesting that the model’s failure mode is “insufficient evidence” rather than “focused on the wrong features.”

3) *Aggregate class-conditional heatmaps*: A potential criticism of per-example Grad-CAM is cherry-picking. To address this we compute the **average** Grad-CAM heatmap across *all* correctly-classified samples of each class (Fig. 10). The aggregates reveal stable, class-specific attention patterns. We compute time-axis statistics (peak location and fraction of energy in the first/middle/last third of the segment) to summarize each pattern quantitatively.

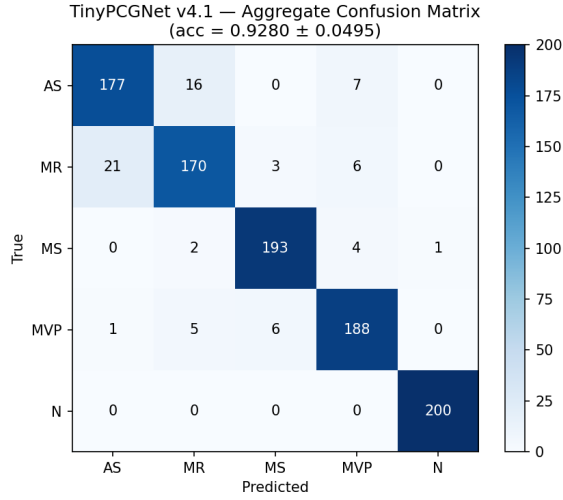


Fig. 7: Aggregate confusion matrix of TinyPCGNet across five folds on the OAHS dataset. Rows are true labels; columns are predictions.

4) *Cardiac-cycle interpretation:* Each 3-second segment contains 3–4 cardiac cycles (assuming typical heart rates of 60–90 bpm), so without explicit cycle segmentation we cannot quantitatively verify that Grad-CAM attention falls within the clinically expected S1/systole/S2/diastole window for each class. What we *can* observe is that aggregate attention is class-distinct and clinically plausible: classes with murmurs known to be holosystolic (MR) show broader temporal coverage, while classes with sharp transients (MVP click) show more concentrated attention. Per-cycle alignment using the Springer segmentation algorithm [10] is a natural extension and represents an active line of recent work [11].

IX. DISCUSSION

A. Architectural insights

The dominance of recurrent layers in our results is striking. A Conv-BiLSTM on MFCC at 100K parameters reaches 99.5% accuracy, while a CNN-2 (otherwise identical conv backbone) on the same MFCC features reaches only 89.9%. The 10-point gap is attributable purely to the recurrent head. This suggests that the temporal axis of MFCC carries substantial class-discriminative information that purely convolutional models lose when they apply Global Average Pooling.

TinyPCGNet, being purely convolutional, also collapses the time axis through GAP. This explains its 92.8% accuracy versus the recurrent models’ 99%+. A natural follow-up is a hybrid TinyPCGNet variant with a lightweight recurrent head on top of each branch — something we leave to future work.

B. Feature insights

Contrary to a common assumption that multi-feature fusion always improves accuracy, our results show that a well-chosen single feature (MFCC) with a well-chosen model (Conv-BiLSTM) beats every fusion model we tested. Fusion remains valuable, however, for parameter-constrained designs

where neither feature alone provides enough discriminative information given the limited model capacity.

C. Edge-deployment implications

Mapping the operating regions identified in Section VIII-B to deployment targets:

- **Smartphone or Raspberry Pi 4** (≥ 1 GB RAM, Cortex-A class): any model in Table I can be deployed. Recommend Conv-BiLSTM $\times$ MFCC for accuracy.
- **Raspberry Pi Zero / Cortex-A53 wearables** (256 MB RAM): keep parameter count under 50K. CNN-2 or TinyPCGNet are good choices.
- **Cortex-M MCU** (e.g., STM32, 256 KB RAM): TinyPCGNet is at the edge of feasibility. INT8 post-training quantization brings the model to roughly 7 KB and likely enables MCU deployment.

X. FUTURE DIRECTIONS

Several promising extensions of this work are worth pursuing:

- 1) **Cross-dataset validation.** The OAHS dataset is clean and balanced; real-world recordings are not. Validation on the PhysioNet/CinC 2016 and 2022 datasets would assess whether these models generalize to noisy, clinically diverse recordings.
- 2) **Real edge benchmark.** Convert TinyPCGNet to TensorFlow Lite INT8 and benchmark on a Raspberry Pi Zero or STM32 MCU. Measure actual latency, RAM footprint, and energy per inference, comparing against the published 1.87 ms latency of WCNN [12] on Raspberry Pi.
- 3) **Cardiac-cycle-aligned Grad-CAM.** Use the Springer segmentation algorithm [10] to label each frame as S1/systole/S2/diastole and quantitatively score how Grad-CAM attention aligns with the clinically expected window per class.
- 4) **Knowledge distillation.** Train a Conv-BiLSTM $\times$ MFCC teacher (99.5%) and distill into TinyPCGNet’s architecture, potentially closing the accuracy gap without increasing parameter count.
- 5) **Noise robustness.** Inject lung sounds and ambient noise at controlled SNRs (5, 15, 25 dB) and measure accuracy degradation across the 22 models. This would reveal whether parameter savings come at a robustness cost.
- 6) **TinyPCGNet with recurrent head.** Add a lightweight BiLSTM or GRU layer on top of each branch before GAP, trading a few thousand parameters for the temporal modeling that clearly benefits the recurrent baselines.

Searching for the accuracy–parameter sweet spot. A central observation of this study is that the parameter–accuracy curve in Section VIII-B exhibits a sharp knee near 5K parameters. However, the precise sweet spot for a given deployment target remains an open question. A natural next step is to treat this trade-off as an explicit optimization problem rather than as a manual architecture-selection exercise.

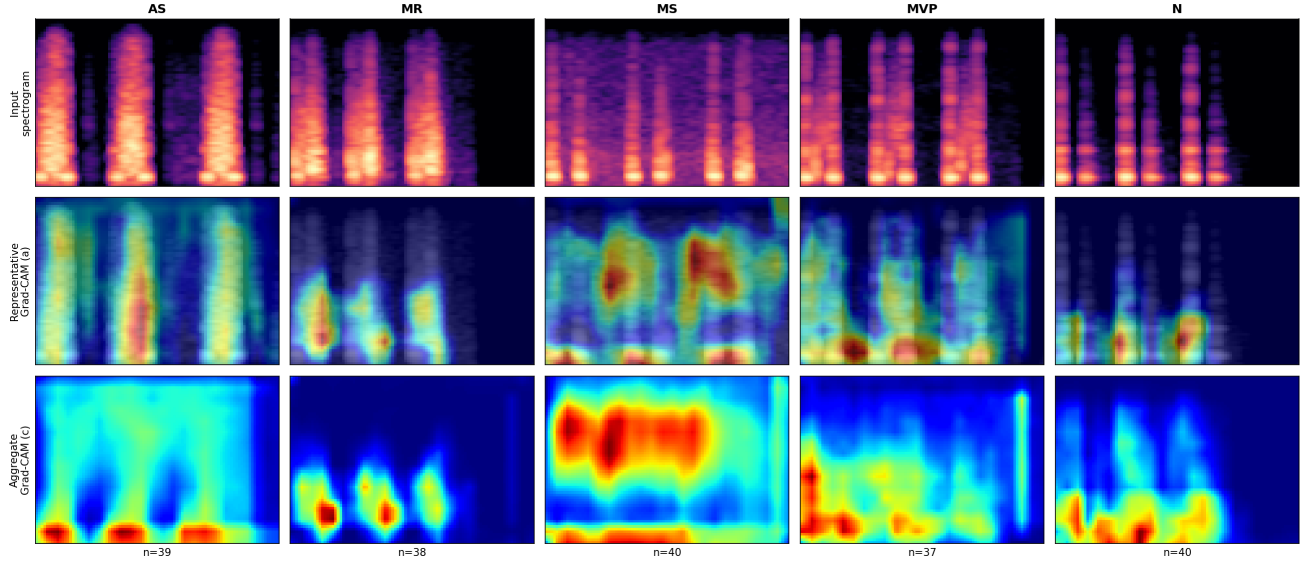


Fig. 8: Per-class representative Grad-CAM. Top row: input log-mel spectrogram. Bottom row: Grad-CAM heatmap overlay. Each column shows one correctly classified example.

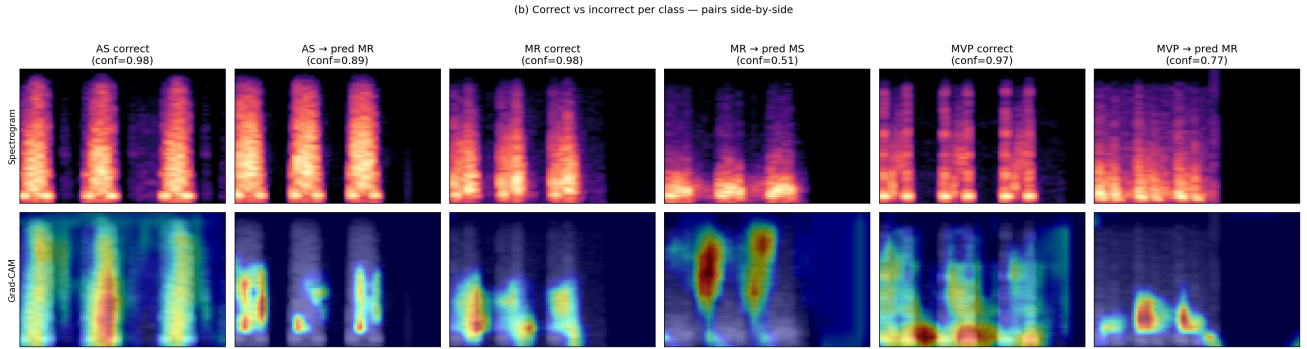


Fig. 9: Correct versus incorrect predictions side-by-side. For each class, the left pair shows a correctly classified example and the right pair shows a misclassification. The misclassified examples typically show more diffuse attention.

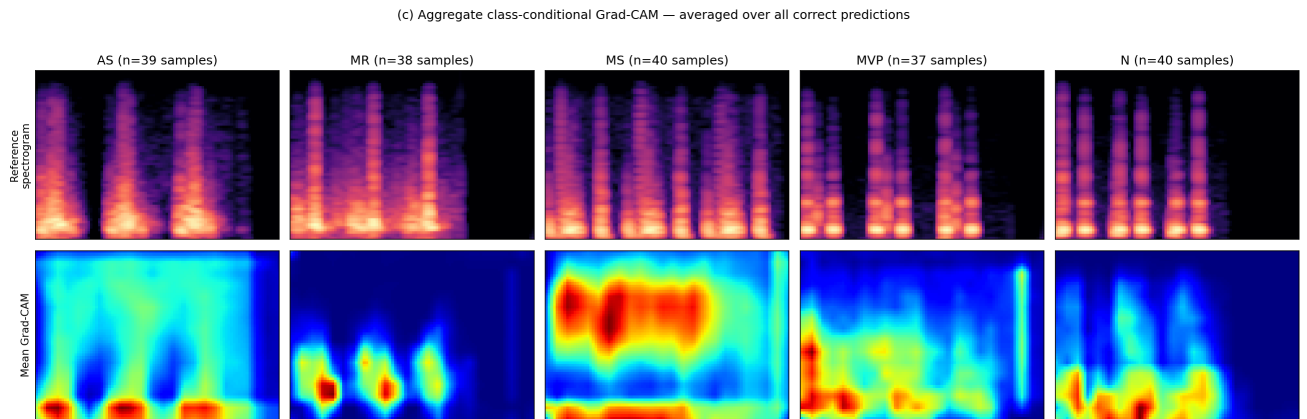


Fig. 10: Aggregate class-conditional Grad-CAM heatmaps averaged across all correctly classified samples of each class. The aggregates reveal stable class-discriminative attention patterns.

Structured pruning of the high-accuracy Conv-BiLSTM×MFCC champion model, which contains approximately 100K parameters and achieves 99.5% accuracy, offers one direct path. By iteratively removing low-magnitude

filters and gating units, followed by fine-tuning the remaining parameters, the full accuracy-degradation curve can be traced. This would allow identification of the smallest model that preserves a target accuracy threshold, such as $\geq 97\%$.

Complementary techniques include *magnitude-based unstructured pruning* for further compression after the structured baseline is established, *N:M semi-structured sparsity* for compatibility with modern accelerators, and *quantization-aware training* to improve upon the post-training INT8 conversion used in this work.

On the architecture side, *lightweight attention mechanisms*, such as efficient channel attention (ECA), coordinate attention, or shifted-window self-attention, could replace the squeeze-and-excitation blocks in TinyPCGNet at comparable or lower parameter cost. This may help close the accuracy gap with recurrent baselines without increasing the parameter budget.

Finally, *Neural Architecture Search* (NAS) with explicit constraints on parameter count and FLOPs could automate this search and produce a Pareto-optimal family of TinyPCGNet variants tailored to specific deployment targets, ranging from a 50K-parameter Raspberry Pi variant to a sub-2K-parameter MCU variant. Together, these directions would transform the single TinyPCGNet operating point reported in this work into a continuous trade-off curve that practitioners can navigate according to their hardware budget.

XI. CONCLUSION

This work presented a systematic comparative study of seven deep architectures across three feature representations for phonocardiogram classification on the OAHS five-class VHD task. By using identical training protocols and fold splits across all 22 experiments, the study enables apples-to-apples architectural comparison absent in the prior literature.

The key findings are: (i) recurrent models on MFCC features dominate the high-accuracy frontier, with Conv-BiLSTM $\times$ MFCC reaching 99.5% accuracy at 100K parameters; (ii) the parameter-accuracy relationship exhibits a clear knee around 5K parameters, identified with our proposed TinyPCGNet model; (iii) TinyPCGNet achieves 92.8% accuracy at 5.1K parameters — a $20\times$ parameter reduction over state-of-the-art LtXe-EnsNet at a 6.7% accuracy cost; and (iv) Grad-CAM analysis reveals class-discriminative attention patterns consistent with the cardiac-cycle structure of each VHD type.

These results characterize the trade-off frontier for PCG classification and demonstrate that ultra-lightweight models remain viable for resource-constrained edge deployment. The choice of operating point on this trade-off curve depends entirely on the target deployment hardware: smartphones can afford the accuracy champions, while microcontroller-class wearables benefit from TinyPCGNet's compact footprint.

REFERENCES

- [1] S. Chowdhury, M. Morshed, and S. A. Fattah, "SpectroCardioNet: An attention-based deep learning network using triple-spectrograms of PCG signal for heart valve disease detection," *IEEE Sensors J.*, vol. 22, no. 23, pp. 22799–22807, Dec. 2022.
- [2] M. Morshed and S. A. Fattah, "SAR-CardioNet: A network for heart valve disease detection from PCG signal based on split-self attention with residual paths," *IEEE Sensors J.*, vol. 23, no. 17, pp. 19553–19560, Sep. 2023.
- [3] M. T. Rahman and C. Shahnaz, "LtXe-EnsNet: A lightweight and explainable ensembled deep learning model for heart sound abnormality classification from sensor data," *IEEE Sensors Letters*, vol. 9, no. 9, Sep. 2025.
- [4] A. Bhardwaj, S. Singh, and D. Joshi, "Explainable deep convolutional neural network for valvular heart diseases classification using PCG signals," *IEEE Trans. Instrum. Meas.*, vol. 72, 2023, Art. no. 2514215.
- [5] C. Divakar, R. Harsha, K. Radha, D. V. Rao, N. Madhavi, and T. Bharadwaj, "Explainable AI for CNN-LSTM network in PCG-based valvular heart disease diagnosis," in *Proc. 14th Int. Conf. Cloud Comput., Data Sci. Eng.*, 2024, pp. 92–97.
- [6] G.-Y. S. Yaseen and S. Kwon, "Classification of heart sound signal using multiple features," *Appl. Sci.*, vol. 8, no. 12, 2018, Art. no. 2344.
- [7] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 7132–7141.
- [8] A. G. Howard *et al.*, "MobileNets: Efficient convolutional neural networks for mobile vision applications," *arXiv preprint arXiv:1704.04861*, 2017.
- [9] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 618–626.
- [10] D. B. Springer, L. Tarassenko, and G. D. Clifford, "Logistic regression-HSMM-based heart sound segmentation," *IEEE Trans. Biomed. Eng.*, vol. 63, no. 4, pp. 822–832, Apr. 2016.
- [11] M. Alrabie and A. Barnawi, "Are AI models listening like cardiologists? Validating clinical alignment of Grad-CAM in PCG classification," *IEEE Access*, vol. 13, 2025 (to appear).
- [12] Y. Author *et al.*, "WCNN: A Grad-CAM-guided lightweight CNN for PCG classification," *MDPI Sensors*, vol. 25, no. 5, 2025.
- [13] C. Liu *et al.*, "An open access database for the evaluation of heart sound algorithms," *Physiol. Meas.*, vol. 37, no. 12, pp. 2181–2213, 2016.